

Morgan Evans

morganfevans@gmail.com • 218-216-5222 • linkedin.com/in/morganfevans

EDUCATION

University of Minnesota, Twin Cities

Minneapolis, MN

College of Science and Engineering,

Bachelor of Aerospace Engineering and Mechanics

TECHNICAL SKILLS

Programming & Software Tools: Python, Matlab, Excel, Google Collab, Microsoft Office Suite

Machine Learning & AI Tools: NumPy, Pandas, Matplotlib, Generative AI

Version Control: Git/Github

Engineering Specific Software: Simulink, Solidworks, AutoCAD, AutoDesk, Inventor, Fusion360, Arduino

Lab Equipment: Rheometer, Instron, Trios Software, Oscilloscope, BlueHill Software, Adhesive: Coating and Lamination

WORK AND RESEARCH EXPERIENCE

Engineer & Technician Aide, 3M Research and Development, Maplewood, MN

January 2025 - Present

- Led and trained technical aides, interns, and new hires while establishing standardized laboratory procedures and documentation practices
- Executed materials characterization and adhesive testing for proprietary R&D projects using Instron systems, rheometers, and microscopy equipment
- Supported experimental design, failure analysis, and process optimization through peel, tensile, viscosity, and dynamic mechanical analysis (DMA) testing
- Automated engineering data analysis and reporting workflows through advanced Excel tool development, reducing analysis time by 80%
- Developed ML/AI-based software solutions supporting proprietary consumer product research and development efforts

Flight Simulation System Architect, Senior Design, University of Minnesota

September 2025 - May 2026

- Developed a physics-based simulation to evaluate control behavior, dynamic response, and operational limits
- Implemented modular configuration files to support requirement driven parametric analysis and design iteration
- Designed time-synchronized data logging and analysis pipelines to assess control stability and system performance
- Architected an AI-based advisory “copilot” that monitors system state and provides decision support guidance based on defined operating criteria

Lead Teaching Assistant, Astrophysics AST 2001, University of Minnesota

September 2023 - May 2026

- Supervised and coordinated 4 TAs, ensuring consistent grading standards and solution development

Lead Research Assistant, AMAAZE Labs, University of Minnesota

2025

- Developed Python code for publication that automated the reorganization of 7,000+ unstructured files, producing a high-quality dataset released as a public resource for use by external researchers and machine learning training

SUPPORTING EXPERIENCE

Guidance, Navigation, and Control Engineer, Rocket Team, University of Minnesota

2022 - 2024

Resident Assistant, Ecumen Seasons, Maplewood, MN

2023 - 2024

Technical Drafting Competitor & CAD Specialist, SkillsUSA, Superior, WI

2018 - 2022

PROFESSIONAL MEMBERSHIPS

Founder & Director, Gift of Life Bone Marrow Donation Registry, University of Minnesota

2023 - 2026

Alpha Sigma Kappa Women in Technical Studies, University of Minnesota

2022 - 2026

American Institute of Aeronautics and Astronautics, University of Minnesota

2022 - 2026